

Multiple Choice (10 marks)

1. Which of the following is an authentication method?
 - (a) Secret question
 - (b) Biometric
 - (c) SMS code
 - (d) All of the above
2. A hash function guarantees the integrity of a message. It guarantees that the message has not be
 - (a) Replaced
 - (b) Overview
 - (c) Changed
 - (d) Violated
3. The AH Protocol provides source authentication and data integrity, but not
 - (a) Integrity
 - (b) Privacy
 - (c) Nonrepudiation
 - (d) Both A & C
4. Let $I = (S, V)$ be a MAC. Suppose $S(k, m)$ is always 5 bits long. Can this MAC be secure?
 - (a) No, an attacker can simply guess the tag for messages
 - (b) It depends on the details of the MAC
 - (c) Yes, the attacker cannot generate a valid tag for any message
 - (d) Yes, the PRG is pseudorandom
5. _____ is the central node of 802.11 wireless operations.
 - (a) WPA
 - (b) Access Point
 - (c) WAP
 - (d) Access Port

True / False (10 marks)

Answer True or False

6. True or False: A session symmetric key can be reused multiple times between the same parties for different sessions without compromising security.
7. True or False: A buffer overflow occurs when the program detects a buffer is full and begins to reject additional data requests.
8. True or False: Weak or non-existent authentication mechanisms are not considered issues associated with the presentation layer.
9. True or False: The message must be encrypted and decrypted at the sender site to maintain data integrity during transmission.
10. True or False: A proxy firewall primarily filters traffic at the physical layer, managing connections and data transmission directly over the network.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the concept of message integrity in cybersecurity, including its significance, methods for ensuring it, and the potential consequences of its compromise.
12. Discuss the role of the session layer in the OSI model and analyze how it can be exploited in brute-force attacks on access credentials.

End of Paper